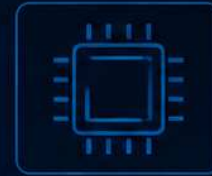


STM32 vs GD32 Middleware Benchmark



VS



Comparative evaluation of middleware efficiency
on Cortex-M33 platforms

DKHILI Fares

Plan

Middleware Stack	Benchmark Scope: ST vs GigaDevice	Headline KPIs	HW Board	Effort estimation
USB Device	Measure USB device stack efficiency across HID, CDC, and MSC through footprint, initialization, enumeration, and transfer performance.	➤ Total footprint ➤ Initialization time ➤ Enumeration time ➤ Transfer latency		• 5 MD
FreeRTOS	Measure RTOS responsiveness and scalability through task, sync, queue, timer, ISR wakeup, and context-switch performance.	➤ Total footprint ➤ Context-switch cost ➤ ISR wakeup latency ➤ API latency	GD32E507Z-EVAL vs STM32U5A9J-DK • Cortex: same M33 • Frequency: 180 MHz vs 160 MHz • Flash: 512 KB vs 4 MB • SRAM: 128 KB vs 2.5 MB	• 5 MD
FatFS	Measure file-system efficiency on external and removable storage through mount, format, and file operation performance.	➤ Total footprint ➤ mount/format time ➤ File operations latency ➤ throughput		• 5 MD
LwIP	Measure Ethernet stack efficiency through startup, network service latency, socket performance, and TCP/UDP throughput.	➤ Total footprint ➤ Bring-up time ➤ TCP/UDP throughput		• 7 MD

(*) estimation based on tests first development will be done by Copilot + Manual configuration adjustment and validation on HW boards

Benchmark Scope: Cube2 STM32C5 vs TI MSPM33	HW Board
• Done before on STM32C0 vs TI MSPM0. Now, we will assess changes in M33 products. • STEP 1 Static analysis: Ecosystem, tools, drivers' architecture, ... will be studied • STEP 2 Functional analysis: • LL equivalent scenarios deployment on STM32C5 and MSPM33 to evaluate footprint • MW equivalent scenarios deployment on STM32C5 and MSPM33 to evaluate footprint and execution time (stacks to be defined)	<u>MSPM33C321A vs STM32C5A3ZJ-NUCLEO</u> • Cortex: same M33 • Frequency: 160 MHz vs 144 MHz • Flash: same 1 MB • SRAM: same 256 KB

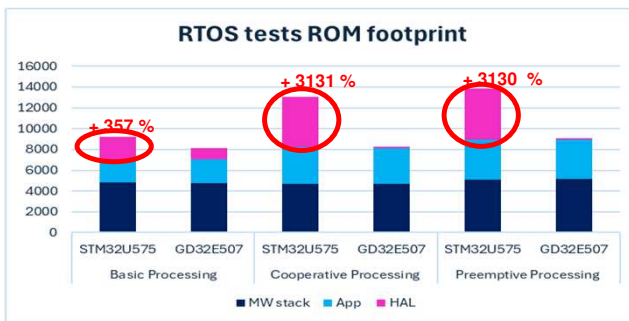


Summary

FreeRTOS tests Footprint

Porting tests from STM32 (existing) to GD32 is performed by Copilot + Manual validation and execution on board

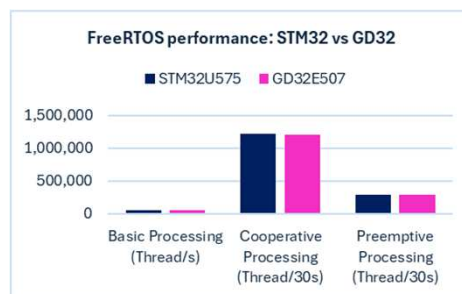
RTOS Scenario	Footprint	STM232 vs GD32
Basic Processing	ROM	+ 40.86 %
	RAM	+2.08 %
Cooperative Processing	ROM	+ 50 %
	RAM	+2.31 %
Preemptive Processing	ROM	+ 46 %
	RAM	+ 2.31 %



- Although STM32 and GD32 have same RAM consumption, but there is a significant gap in ROM footprint.
- ROM footprint splitting shows that STM32 and GD32 tests have same Application footprint (same scenario) and FreeRTOS footprint (same stack), however the STM32 HAL drivers' footprint is much higher than GD32, mainly due to STM32 HAL RCC driver footprint

FreeRTOS tests performance (thread/s)

RTOS Scenario	STM232 vs GD32
Basic Processing (Thread/s)	- 0.025 %
Cooperative Processing (Thread/30s)	+ 1.6 %
Preemptive Processing (Thread/30s)	+ 3.3 %

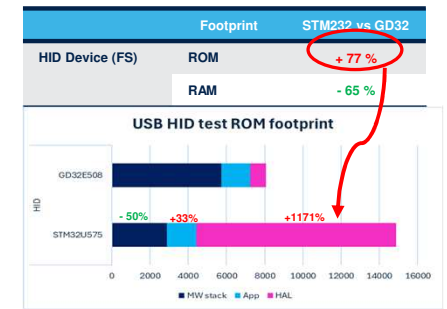
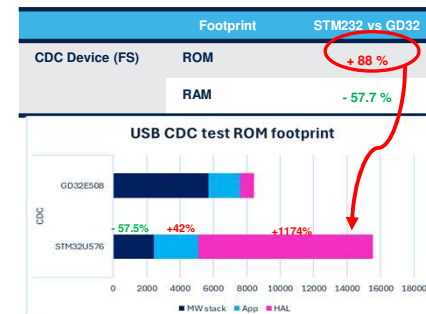


STM32 and GD32 FreeRTOS tests have almost the same performance



STM32/GD32 USB Library Footprint

Porting tests from STM32 (existing) to GD32 is performed by Copilot + Manual validation and execution on board



- STM32 ROM footprint is **higher by 88% in CDC Device and 77% in HID Device** than GD32. This is due to:
 - Application ROM:** STM32 **higher by 42% in CDC Device/ 33% in HID Device** than GD32 because of USB Device user files: usb_device.c, usbd_cdc_if.c, usbd_desc.c, usbd_conf.c (43% in CDC Device/38% in HID Device of total App ROM footprint).
 - In GD32, USB device configuration is managed directly in MW stack
 - HAL ROM:** STM32 **higher by average 1172%** than GD32, mainly due to:
 - HAL RCC: 4560 bytes => 44% of total HAL ROM footprint
 - HAL/LL USB: 5006 bytes => 48% of total HAL ROM footprint
- The high STM32 App and HAL ROM footprint lead to a **lower STM32 USB library ROM footprint** than GD32 USB library by 57.5% in CDC Device/50% in HID Device

STM32/GD32 USB Library Execution time (ms)

USB CDC operation	STM232 vs GD32
Initialization	-61.68%
Enumeration	-0.14%
Transmit data	-1.53%
Receive data	-0.72%

USB HID operation	STM232 vs GD32
Initialization	-61.70%
Enumeration	-1.39%
Data transmission	-0.71%

STM32 USB Device **outperforms GD32 by 61%** in initialization phase.

STM32 and GD32 USB Device tests have almost the same execution time in Enumeration, data transmission and reception phases.

STM32 vs GD32 MW Benchmark

Plan

Introduction



Defined the benchmark scope, KPIs, and comparison methodology.

01

FreeRTOS



Compared normalized native FreeRTOS scenarios on STM32 and GD32.

02

USB Stack



Benchmarked HID, CDC across footprint, timing, and transfer metrics.

03

Benchmark support tools

SE1



Developed custom tools for map analysis, cross-vendor comparison, and AI-assisted support.

04

Conclusion



Summarized results and continuation plan

05

Slide 4

SE1 Better to name it: AI-based footprint analyzer tool

Sirine EL JAZI, 2026-06-15T14:31:04.349

FD1 0 I have developped 2 things, the first is the ai based footprint analyzer tool just for footprint , then i developed a custom copilot agent which create port, build and debug projects

Fares DKHILI, 2026-06-15T15:25:30.614

FD1 1 I will name it benchmark support tool , I think it's better

Fares DKHILI, 2026-06-15T15:26:43.692

STM32 vs GD32 MW Benchmark

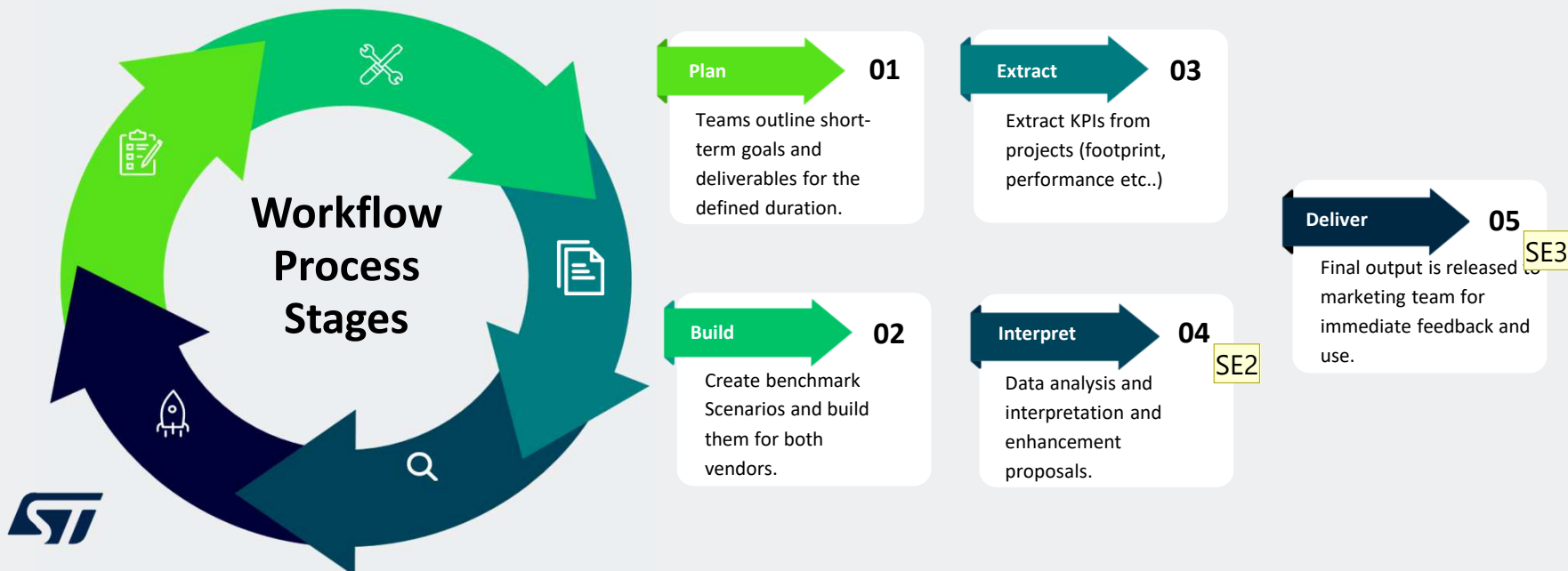
Introduction



Objective

- ❑ Develop similar scenarios on STM32 and GD32 platforms using their embedded software frameworks.
- ❑ Conduct functional testing to assess performance, memory footprint and robustness
- ❑ Compile a comprehensive benchmark report summarizing the results, insights, and recommendations for ST framework improvements

SE1



Slide 5

SE1 Develop similar scenarios on STM32 and GD32 platforms using their embedded software frameworks.

Sirine EL JAZI, 2026-06-15T14:33:36.389

SE2 Check this number

Sirine EL JAZI, 2026-06-15T14:34:57.353

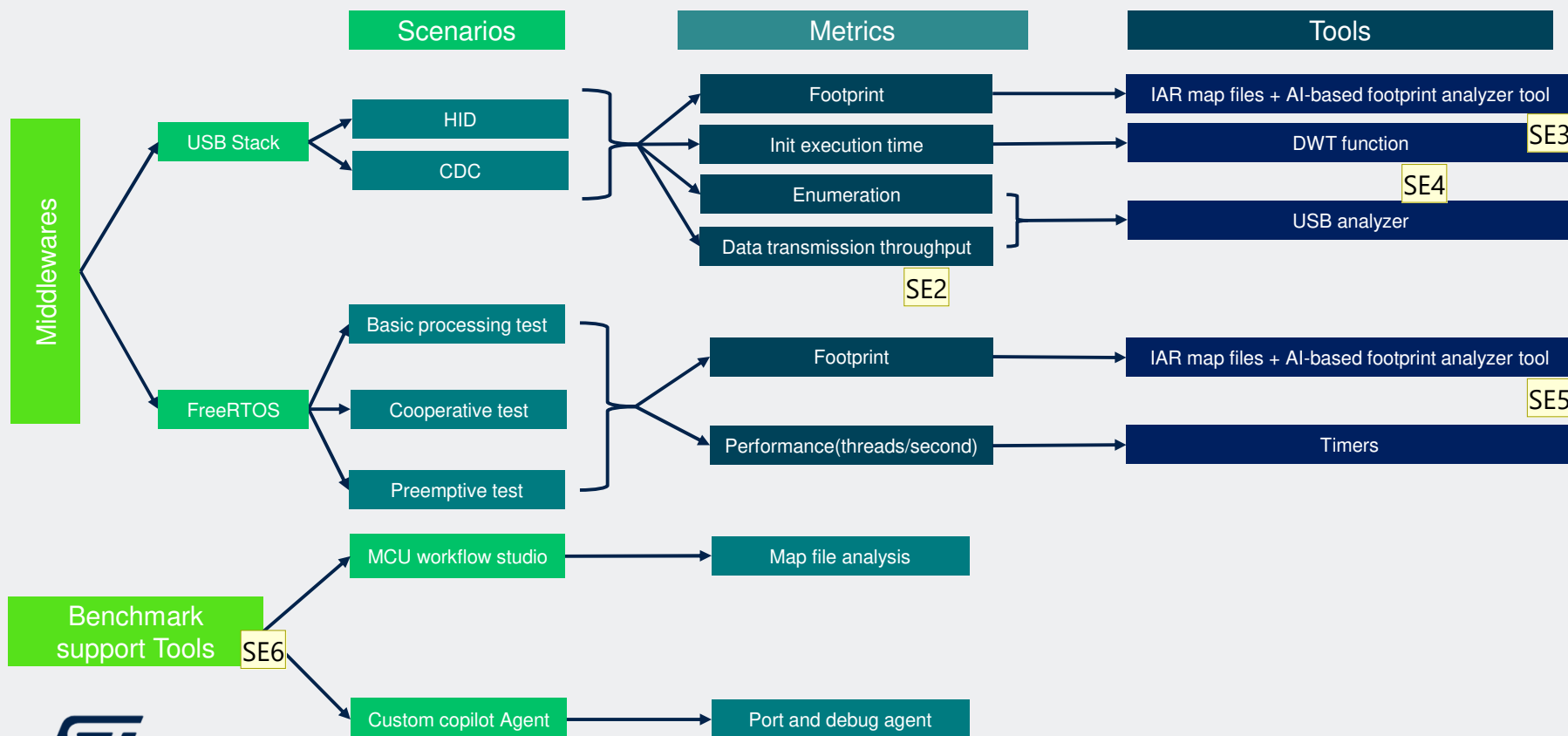
SE3 Check this number

Sirine EL JAZI, 2026-06-15T14:35:10.628

STM32 vs GD32 MW Benchmark

Done work

Benchmarking middleware's behavior across **STMicroelectronics** and **Giga Device**



Slide 6

- SE1** To be removed
Sirine EL JAZI, 2026-06-15T14:35:38.870
- SE2** Data transmission throughput
Sirine EL JAZI, 2026-06-15T14:36:13.824
- SE3** AI-based footprint analyzer tool
Sirine EL JAZI, 2026-06-15T14:36:48.498
- SE4** method
Sirine EL JAZI, 2026-06-15T14:37:03.931
- SE5** AI-based footprint analyzer tool
Sirine EL JAZI, 2026-06-15T14:37:27.903
- SE6** AI-based footprint analyzer tool
Sirine EL JAZI, 2026-06-15T14:37:52.215